

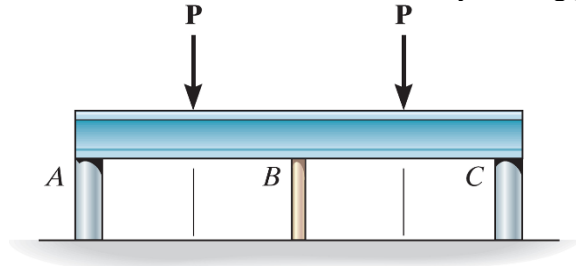
Solid Mechanics (EMCH260 Section 002, Fall 2021)

Mid-Term Exam 2, 8:30 AM, Monday, Oct. 18th 2021

Name (First/Last): _____ Score: _____/100'

1. Single choice questions. There is one and only one correct answer for each question. ($10 \times 3' = 30'$ in total)

Q1: The rigid beam is supported by three posts. If we need to know the stress in each post, is this a determinate or indeterminate uniaxially loading problem?

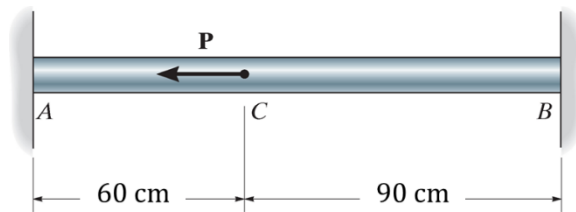


- A. Determinate
- B. Indeterminate

Answer: B

Explanation: There are three unknown reactions but you can only propose two equations (one for total force and the other for total moment). So, you would need an additional equation to solve it.

Q2: The bar is fixed at two ends and the axial load P is applied to C . What compatibility condition can you propose to solve this problem?



- A. $\delta_{AC} = \delta_{CB}$
- B. $\delta_{AC} + \delta_{CB} = 0$

Answer: B

Explanation: Upon loading, the net displacement between two ends A and B should be zero, which can be broken down into two parts, i.e. $\delta_{AB} = \delta_{AC} + \delta_{CB} = 0$. The choice A is not correct, since $\delta_{AC} = -\delta_{CB}$.

Q3: What type of stress is developed in the cross-section perpendicular to the longitudinal axis of a shaft when external torque is applied?

- A. Normal stress
- B. Shear stress

Answer: B

Explanation: The torsion formula $\tau = \frac{T\rho}{J}$ defines the shear stress developed in the cross-section perpendicular to the longitudinal axis of a shaft when external torque is applied. See page 7 in the PPT of Chapter 5.1.

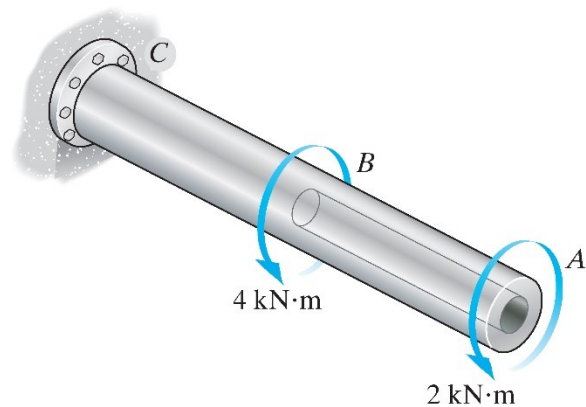
Q4: If a solid shaft has the same diameter as the outer diameter of a hollow or tubular shaft, which one has higher polar moment of inertia?

- A. The solid shaft
- B. The hollow or tubular shaft

Answer: A

Explanation: The polar moment of inertia for a solid circular cross-section is $J = \frac{\pi}{2} c^4$, while the polar moment of inertia for a hollow cross-section is $J = \frac{\pi}{2} (c_o^4 - c_i^4)$. When the solid shaft has the same diameter as the outer diameter of a hollow or tubular shaft, i.e. $c = c_o$, obviously the solid shaft will have higher polar moment of inertia. Refer to page 8 in the PPT of Chapter 5.1.

Q5: The shaft is hollow from *A* to *B* and solid from *B* to *C*. The entire shaft is fixed to the wall at end *C*. If we need to determine the reaction at end *C*, is this a determinate or indeterminate torsional problem?

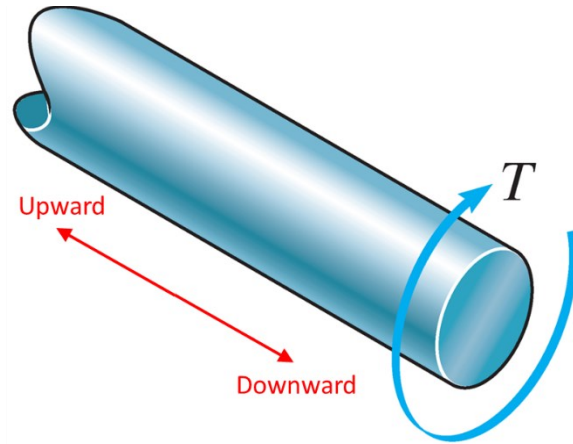


- A. Determinate
- B. Indeterminate

Answer: A

Explanation: We can simply determine the reaction at end C by using Free-Body-Diagram to the entire shaft. So, this problem is determinate.

Q6: Using the right-hand-rule, determine the direction the external torque T is pointing to:

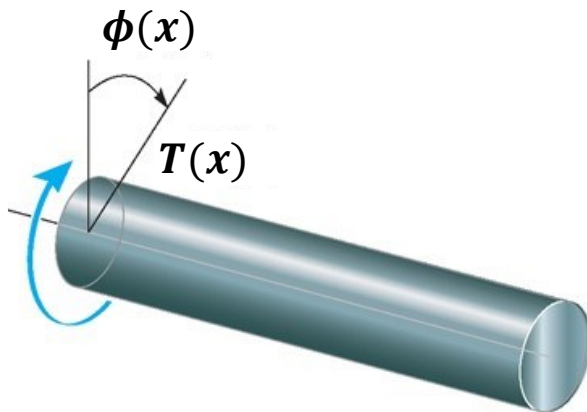


- A. Upward
- B. Downward

Answer: A

Explanation: The thumb is directed upward when the fingers of your right hand curl in the direction of the torque. Refer to page 8 in the PPT of Chapter 5.2.

Q7: Using the right-hand-rule and sign convention, the sign of the internal resultant torque $T(x)$ is:



- A. Negative
- B. Positive

Answer: B

Explanation: Both the torque and twist angle will be *positive*, provided the thumb is directed *outward* from the shaft while the fingers curl in the direction of the torque. Refer to page 8 in the PPT of Chapter 5.2.

Q8: What type of stress is involved in the flexure formula for a bending problem?

- A. Normal stress
- B. Shear stress

Answer: A

Explanation: The normal stress is involved in the flexure formula $\sigma = -\frac{My}{I}$. Refer to page 10 in the PPT of Chapter 6.3.

Q9: Use beam sign convention to determine the sign of the shear force shown below (the blue region is a segment of the beam):

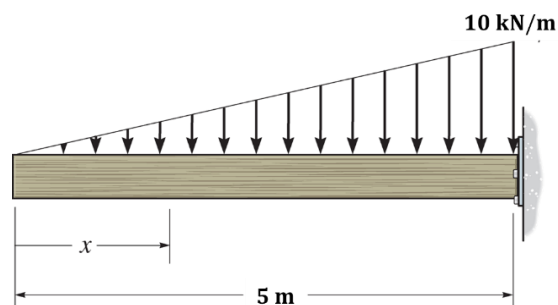


- A. Negative
- B. Positive

Answer: B

Explanation: The positive sign is defined for shear force if the V lets the segment rotate *clockwise*. Refer to page 7 in the PPT of Chapter 6.1.

Q10: Determine the total equivalent force of the external distributed forces:

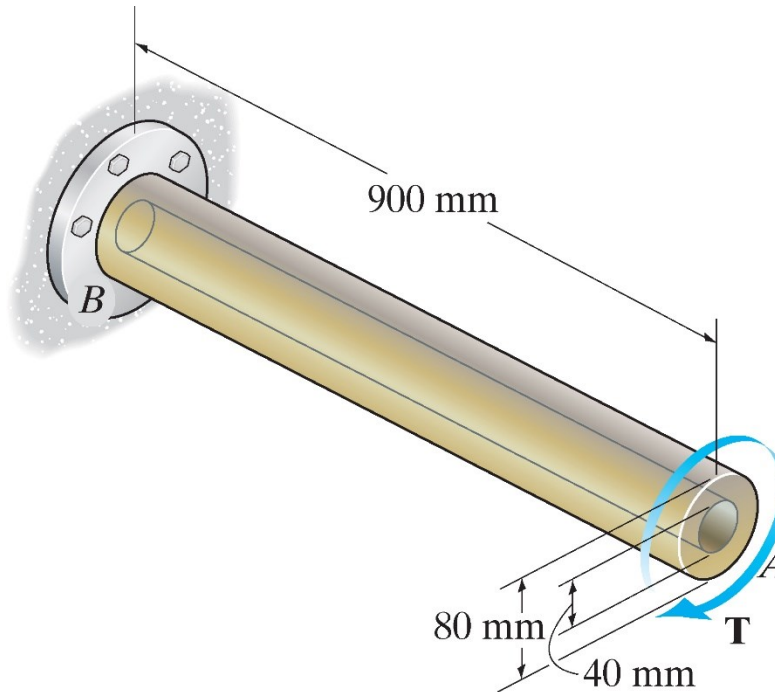


- A. 25 kN
- B. 50 kN

Answer: A

Explanation: The total equivalent force of the distributed forces can be calculated by the area of the triangle, i.e. $10 \frac{\text{kN}}{\text{m}} \times 5 \text{ m} / 2 = 25 \text{ kN}$.

2. The magnesium tube is bonded to the steel rod. If the allowable shear stresses for the magnesium and steel are $(\tau_{allow})_{mg} = 45 \text{ MPa}$ and $(\tau_{allow})_{st} = 75 \text{ MPa}$, respectively, determine the maximum allowable torque that can be applied at end A . Also, find the corresponding angle of twist of the end A . Take the shear modulus $G_{mg} = 20 \text{ GPa}$ (magnesium) and $G_{st} = 80 \text{ GPa}$ (steel). (30' in total)



Solutions:

Step 1: Equilibrium

Assuming the resultant internal torques in the magnesium tube and steel rod as T_{mg} and T_{st} , respectively. For the entire specimen, the free-body diagram (moment equation of equilibrium) gives:

$$\sum M = 0 \rightarrow$$

$$T_{mg} + T_{st} = T \quad (1)$$

Step 2: Compatibility Condition

Since the steel rod is bonded firmly to the magnesium tube, the angle of twist of the rod and the tube must be the same, i.e.

$$\phi_{mg} = \phi_{st} \quad (2)$$

Step 3: Load-Displacement Relations

The angle of twist for the shaft and the tube can be obtained by the internal torque, shear modulus of elasticity (note that the magnesium tube and steel rod have different shear modulus), and the polar moment of inertia:

$$\phi_{mg} = \frac{T_{mg} L_{AB}}{J_{mg} G_{mg}}$$

$$\phi_{st} = \frac{T_{st} L_{AB}}{J_{st} G_{st}}$$

Substituting them into Eq. (2) and considering $J_{mg} = \frac{\pi}{2}(0.04^4 - 0.02^4)$, $J_{st} = \frac{\pi}{2} \times 0.02^4$, $G_{mg} = 20 \text{ GPa}$, $G_{st} = 80 \text{ GPa}$, we get

$$T_t = 39.85 T_s \quad (3)$$

Combining Eqs. (1) and (3), we get

$$T_{mg} = 0.79 T$$

$$T_{st} = 0.21 T$$

Step 4: Allowable Shear Stress

$$(\tau_{allow})_{mg} = \frac{T_{mg} c_2}{J_{mg}}$$

$$\rightarrow 45 \times 10^6 = \frac{0.79 T \times 0.04}{\frac{\pi}{2}(0.04^4 - 0.02^4)} \quad \rightarrow T = 5368.5 \text{ N} \cdot \text{m}$$

$$(\tau_{allow})_{st} = \frac{T_{st} c_1}{J_{st}}$$

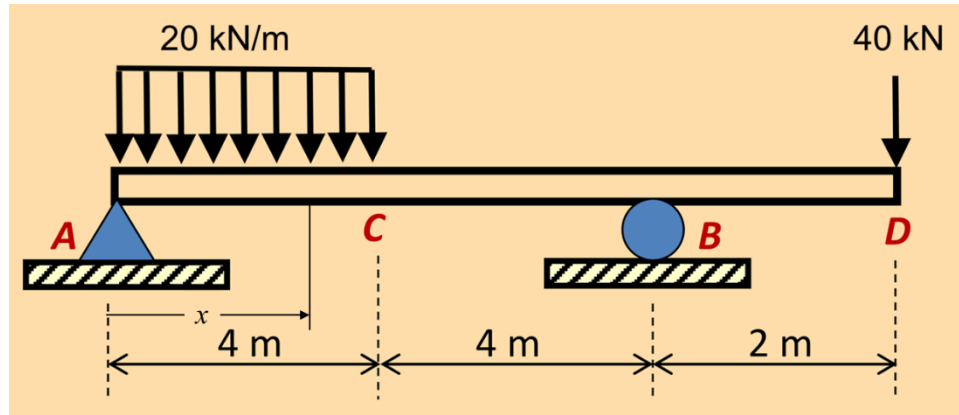
$$\rightarrow 75 \times 10^6 = \frac{0.21 T \times 0.02}{\frac{\pi}{2} \times 0.02^4} \quad \rightarrow T = 4488.0 \text{ N} \cdot \text{m}$$

Thus, the maximum allowable torque that can be applied at the end A is **4488 N·m**

The angle of twist of end A can be obtained by either component in the specimen:

$$\phi_A = \phi_{st} = \frac{T_{st} L_{AB}}{J_{st} G_{st}} = \frac{0.21 \times 4488 \times 0.9}{\frac{\pi}{2} \times 0.02^4 \times 80 \times 10^9} = \mathbf{0.042 \text{ rad}}$$

3. Determine the shear and moment diagrams for the beam and plot the diagrams. Also determine the absolute maximum normal stress in the beam with external loading shown below. The beam has a uniform square cross-section with lateral size $a = 0.2$ m. Note: the shear and moment diagrams can be calculated either by section method or graphical method. (40' in total)



Solutions:

Step 1: Support Reactions

Applying free-body diagram to the entire beam:

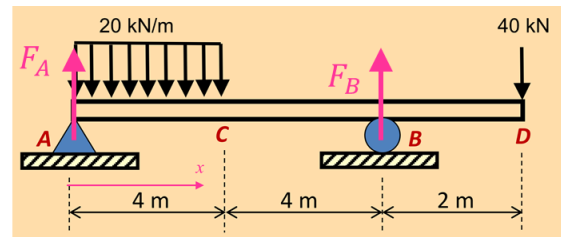
$$\sum F_y = 0 \rightarrow F_A + F_B = 20000 \times 4 + 40000 = 120000 \quad (1)$$

$$\sum M_A = 0 \rightarrow F_B \times 8 = 20000 \times 4 \times 2 + 40000 \times 10 \quad (2)$$

Solving Eqs. (1) and (2) we get:

$$F_A = 50000 \text{ N}$$

$$F_B = 70000 \text{ N}$$



Step 2: Shear Diagram

a) Determine the shear at left end first (the origin A of x-axis):

$$x = 0, V = F_A = 50000 \text{ N}$$

b) Determine V diagram according to $\frac{dV}{dx} = w$ for AC region:

$$\frac{dV}{dx} = -20000$$

$$\rightarrow V = -20000x + C_1$$

The constant C_1 can be determined by applying the “**boundary**” condition ($x = 0$, $V = 50000$). Thus, $C_1 = 50000$ and $V = -20000x + 50000$ ($0 \leq x \leq 4$).

c) Determine **V** diagram according to $\frac{dV}{dx} = w$ for CB region:

$$\frac{dV}{dx} = 0$$

$$\rightarrow V = C_2$$

The constant C_2 can be determined by applying the “**continuity**” condition at last point C ($x = 4$, $V = -30000$). Thus, $C_2 = -30000$ and $V = -30000$ ($4 \leq x \leq 8$).

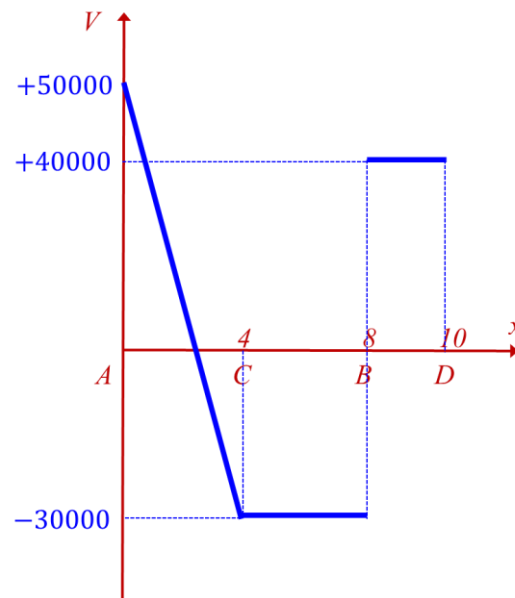
d) At point B, due to the supporting reaction $F_B = 70000$ N (upward), the **V** increases (upward jump) by 70000 N. Then, the **V** jumps from -30000 to $+40000$ N. See the shear diagram for details.

e) Determine **V** diagram according to $\frac{dV}{dx} = w$ for BD region:

$$\frac{dV}{dx} = 0$$

$$\rightarrow V = C_3$$

The constant C_3 can be determined by the “**continuity**” condition at last point B after considering the jump ($x = 8$, $V = 40000$). Thus, $C_3 = 40000$ and $V = 40000$ ($8 \leq x \leq 10$).



Step 3: Moment Diagram

a) Determine the moment at left end first (the origin A of x-axis):

$x = 0$, $M = 0$ (since there is no external bending moment and bending reaction at point A)

b) Determine **M** diagram according to $\frac{dM}{dx} = V$ for AC region:

$$\frac{dM}{dx} = V = -20000x + 50000$$

$$\rightarrow M = -10000x^2 + 50000x + C_4$$

The constant C_4 can be determined by applying the “**boundary**” condition ($x = 0$, $M = 0$). Thus, $C_4 = 0$ and $M = -10000x^2 + 50000x$ ($0 \leq x \leq 4$).

c) Determine **M** diagram according to $\frac{dM}{dx} = V$ for CB region:

$$\frac{dM}{dx} = V = -30000$$

$$\rightarrow M = -30000x + C_5$$

The constant C_5 can be determined by applying the “**continuity**” condition at last point C ($x = 4$, $M = 40000$). Thus, $C_5 = 160000$ and $M = -30000x + 160000$ ($4 \leq x \leq 8$).

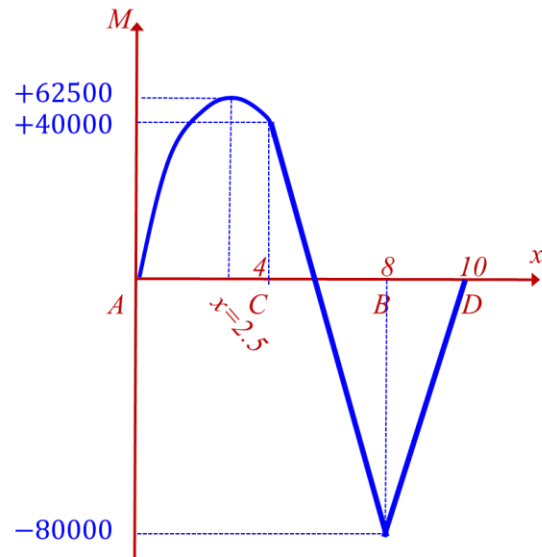
d) Determine **M** diagram according to $\frac{dM}{dx} = V$ for BD region:

$$\frac{dM}{dx} = V = 40000$$

$$\rightarrow M = 40000x + C_6$$

The constant C_6 can be determined by the “**continuity**” condition at last point B ($x = 8$, $M = -80000$). Thus, $C_6 = -480000$ and $M = 40000x - 480000$ ($8 \leq x \leq 10$).

From the moment diagram, we can find the maximum moment occurs at $x = 8$ m, i.e. $M_{\max} = 80000 \text{ N} \cdot \text{m}$



Step 4: Section Property

The moment of inertia I of sectional area about the neutral axis is (refer to the inside cover of the textbook for rectangular cross-section):

$$I = \frac{1}{12}bh^3 = \frac{1}{12} \times 0.2 \times 0.2^3 = 1.3333 \times 10^{-4} \text{ m}^4$$

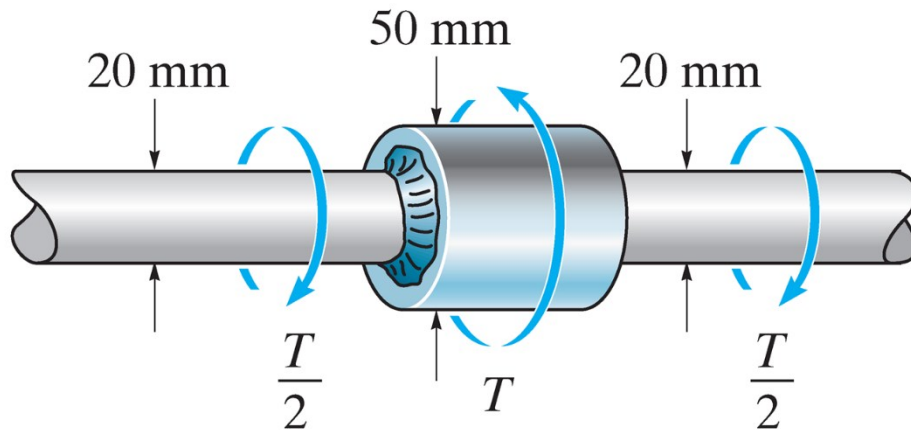
Step 5: Maximum Normal Stress

The maximum normal stress is determined by:

$$\sigma_{\max} = \frac{M_{\max}c}{I}$$

$$\text{Thus, } \sigma_{\max} = \frac{80000 \times 0.1}{1.3333 \times 10^{-4}} = \boxed{60 \text{ MPa}}$$

***[**Bonus Problem**] A solid step shaft has different cross section size (see figure). An external torque $T = 500 \text{ N} \cdot \text{m}$ is applied. Determine the maximum shear stress developed in the shaft. Neglect the transition region between the cross sections. (**10'** in total)



Solutions:

According to the torsion formula, for both thin and thick cross sections, the maximum shear stress occurs at the surface of the shaft.

For small diameter region:

$$\tau_{max}|_1 = \frac{T/2 \times r_1}{J_1} = \frac{500/2 \times 0.01}{\frac{\pi}{2} \times 0.01^4} = 159.15 \text{ MPa}$$

For large diameter region:

$$\tau_{max}|_2 = \frac{T \times r_2}{J_2} = \frac{500 \times 0.025}{\frac{\pi}{2} \times 0.025^4} = 20.37 \text{ MPa}$$

Comparing the two regions we know that the maximum shear stress developed in the shaft is

159.15 MPa